



Research article

Real-time tracking control of electro-hydraulic force servo systems using offline feedback control and adaptive control



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ABSTRACT

This paper focuses on an application of an electro-hydraulic force tracking controller combined with an offline designed feedback controller (ODFC) and an online adaptive compensator in order to improve force tracking performance of an electro-hydraulic force servo system (EHFS). A proportional-integral controller has been employed and a parameter-based force closed-loop transfer function of the EHFS is identified by a continuous system identification algorithm. By taking the identified system model as a nominal plant model, an H_∞ offline design method is employed to establish an optimized feedback controller with consideration of the performance, control efforts, and robustness of the EHFS. In order to overcome the disadvantage of the offline designed controller and cope with the varying dynamics of the EHFS, an online adaptive compensator with a normalized least-mean-square algorithm is cascaded to the force closed-loop system of the EHFS compensated by the ODFC. Some comparative experiments are carried out on a real-time EHFS using an xPC rapid prototype technology, and the proposed controller yields a better force tracking performance improvement.

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1. Introduction

Electro-hydraulic servo systems play an irreplaceable role in modern industry since it combines many merits of fast response, distinguished force to mass ratio, large stiffness, and excellent load capability, and therefore investigation of the electro-hydraulic servo systems is of great interest in both academia and industry [1,2]. An electro-hydraulic force servo system (EHFS) as a typical implementation of the electro-hydraulic technology is employed to artificially recreate complex dynamic loading conditions in laboratory so as to simulate the force interaction with environment [3], and hence the EHFS possesses extensive engineering applications including the automotive industry [4,5], seismic engineering [1], the aerospace industry [6,7], robotics [8], and so on. However, electro-hydraulic servo systems are inherent nonlinear systems [9] due to the presence of servo valve dynamics, especially servo valve

stiction performance [10–12], oil flow, spherical joint clearance, flow-pressure characteristics [13], and so on, and there exists many uncertainties and disturbances in the EHFS system, which results in a challenging problem for the high precision force tracking performance [3]. The servo valve is a key component in the EHFS; however, its stiction performance widely exists in the servo valve and it results in limit cycles in the EHFS, which affects stability of the force closed-loop system [10–12]. A data-driven model for control valve stiction was established in [10]. With consideration of the presence of external disturbances in control valves, different models and identification methods were carried out in [11]. In [12], modeling, detection/quantification, and compensation of control valve stiction were reviewed and systematized. Robust control based on feedback control is widely employed in the EHFS and it can yield satisfactory performance in the presence of system uncertainties, variations, and disturbances [13,14]. In [15], a force controller designed by a quantitative feedback theory (QFT) for an aerodynamic load simulator was presented, and the effectiveness of the QFT controller was verified by numerical simulation. In [4], a QFT controller for the force control of an electro-hydraulic road simulator under assumed parameter uncertainties was employed and the obtained controller was a low-order system suitable for real-time systems.

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In [16], a robust adaptive controller was developed, in which a back propagation algorithm was applied to online tune QFT parameters so that the controller can be adapted under a wide range of working conditions. In [17], a robust force controller was proposed for a novel electro-hydraulic structure in the presence of a polytopic uncertainty. Other methods have been attempted to improve the force tracking performance of the EHFS. Nonlinear backstepping approaches were discussed by many researchers to recursively design a controller with some state variables as virtual intermediate control so as to form the final control laws, and Lyapunov stability for the whole system is guaranteed. In [18], a nonlinear system with friction compensation was analyzed and a Lyapunov function that provided exponentially stable force tracking accuracy was derived for an electro-hydraulic servo system. In [19], an observer-based backstepping force control scheme was developed for a single-rod electro-hydraulic actuator, and a proportional-integral (PI) observer was employed to estimate system states as only the force output was available. In [20], an adaptive robust backstepping controller was constructed for a hydraulic load simulator in the presence of external disturbances, in which a stable parameter adaptation was combined to eliminate the effect of unknown but constant parameter uncertainties.

In contrast to offline designed controllers, online adaptive controllers that can cope with variations and uncertainties of electro-hydraulic servo systems are an effective alternative for the EHFS. In [21], an online recursive matrix factorization algorithm was employed to adaptively adjust varying parameters of an electro-hydraulic force servo system, and the feasibility of the adaptive controller was verified by experiments. In [22], an integrated control strategy was presented for an electro-hydraulic force loading system, where an online adaptive filtered-x least-mean-square (LMS) algorithm was incorporated with a feed-forward inverse controller in order to improve the force tracking performance. In [23], an adaptive nonlinear optimal compensation scheme was proposed to minimize the force tracking error using online estimation of nonlinear parameters of an electro-hydraulic loading system.

Several compound control methods for the EHFS have been proposed to improve the force tracking performance. In [24], a compound control strategy, which combined velocity compensation with an active disturbance rejection controller, was designed for an electro-hydraulic servo control system. In [25], a nonlinear robust dual-loop control scheme that combined a velocity feed-forward compensator with a robust torque controller was proposed to suppress motion disturbance of a hydraulic actuator. In [3], a combined feed-forward force controller, including a modified force inverse model compensator and a velocity feed-forward compensator with internal model control, was proposed to compensate surplus force disturbance caused by active motion of an electro-hydraulic shaking table and yield a high-fidelity force loading tracking performance. In [26], a grey prediction model and a fuzzy proportional-integral-derivative (PID) controller were proposed to eliminate the disturbance force of a hydraulic load simulator. In [27], a feed-forward observer controller combined with a robust feedback controller were presented to reduce the disturbance force of an electro-hydraulic control loading system and its stability was proved.

Despite the existence of the above mentioned control algorithms, high-accuracy force tracking for electro-hydraulic servo systems is challenging and requires complicated control algorithms used even for simple force control tasks [14]. The main contribution of this paper is to design a novel controller using an offline designed feedback controller (ODFC) and an online adaptive compensator, and its implementation in an experimental EHFS. Firstly, a force closed-loop system of the EHFS is controlled by a conventional PI controller. Next, in order to enhance the force

tracking performance, an optimized feedback controller is offline designed by the H_∞ theory on the basis of the PI-controlled EHFS, in which the system performance weighting function, the input weighting function, and the complementary sensitivity weighting function are properly selected in advance. Finally, to overcome the disadvantage of the offline designed controller, especially in the presence of time-varying parameters, an online compensator employing a normalized least-mean-square (nLMS) algorithm is integrated into the EHFS, and variations and uncertainties of the EHFS can be further reduced to obtain a better force tracking performance improvement.

This paper consists of five sections organized as follows. In Section 2, an experimental EHFS test rig is set up; its dynamic model is established and compared with the experiment result. In Section 3, the design procedure of the proposed controller combining the ODFC and the online adaptive compensator is presented and discussed. In Section 4, an experimental control system with the proposed controller is designed and some comparative experimental results are carried out on an experimental EHFS for the proposed controller. Main conclusions are presented in Section 5.

2. Experimental EHFS and its dynamic model

2.1. Experimental EHFS

An experimental EHFS shown in Fig. 1 is employed for experimental performance evaluation of different controllers. The experimental EHFS consists of a $0.8 \text{ m} \times 0.8 \text{ m}$ platform mounted on a base with a horizontal reaction wall using two linear guides and two linear spherical hinges, a servo valve controlled hydraulic actuator with a 70 mm bore and a 50 mm rod providing a force loading input, a servo valve (G761-3004) manufactured by Moog, Inc with 38 L/min flow capacity at 7 MPa supply pressure, specimen, a real-time computer controller, sensors, a hydraulic power supply system, and a control system including hardware and software. A linear variable differential transformer (LVDT) is attached to the hydraulic actuator to real-time measure its displacement and a force sensor is fixed between the hydraulic actuator and the spherical hinge to measure the present real-time

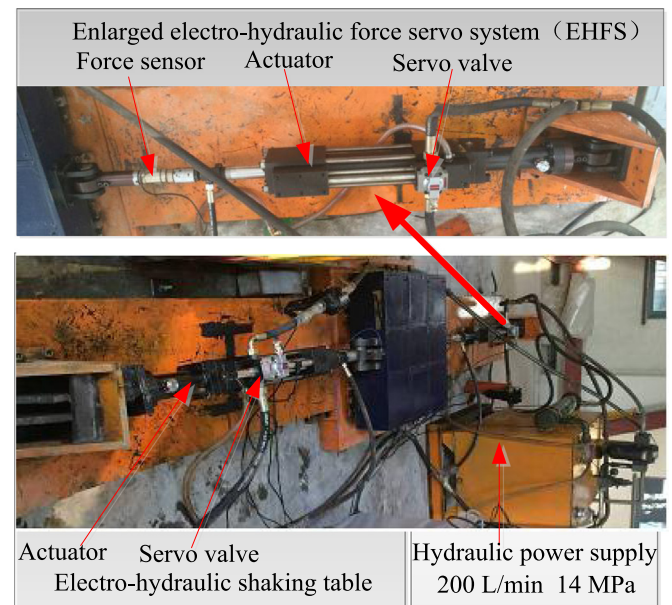


Fig. 1. Experimental EHFS.

Table 1
Main parameters of the EHFS.

Parameters	Values
Platform size	0.8 m×0.8 m
Control type	Force and position
Range of work frequency	0–20 Hz
Actuator stroke	± 100 mm
Maximum velocity	0.2 m/s
Maximum payload	5000 N
Range of force loading	± 15,000 N

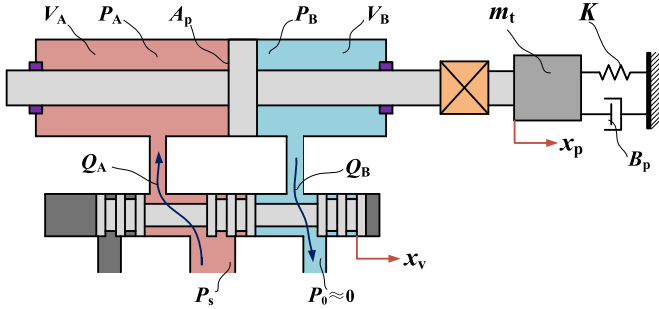


Fig. 2. Simplified working principle of the EHFS.

force feedback signal. The specific system parameters are listed in Table 1.

2.2. EHFS dynamic model

A simplified actuating configuration structure of the EHFS includes a platform, a servo valve, a hydraulic cylinder and its power supply, and force and displacement sensors; their relationship is demonstrated in Fig. 2.

The linearized load flow Q_L of the servo valve can be described by

$$Q_L = K_q x_v - K_c P_L, \quad (1)$$

where x_v is the spool displacement of the servo valve, P_L is the pressure drop across the load defined by $P_L = P_A - P_B$, K_q and K_c are the linearized flow gain and flow pressure coefficient, respectively, which can be obtained by differentiating the accurate load flow Q_L , expressed as [28]

$$K_q = \frac{\partial Q_L}{\partial x_v} = C_d w_v \sqrt{\frac{P_s - \frac{x_v}{|x_v|} P_L}{\rho}}, \quad (2)$$

$$K_c = -\frac{\partial Q_L}{\partial P_L} = C_d w_v |x_v| \sqrt{\frac{1}{4\rho \left(P_s - \frac{x_v}{|x_v|} P_L \right)}}, \quad (3)$$

where C_d is the discharge coefficient, w_v is the constant area gradient of valve orifices, ρ is the hydraulic oil density, and P_s is the supply pressure. With consideration of the oil compressibility, internal leakage, external leakage, and volume changes of the two chambers, the flow continuity equation for every chamber of the cylinder can be expressed by [29]

$$\begin{cases} Q_A = A_p \frac{dx_p}{dt} + C_{ip} P_L + C_{ep} P_A + \frac{V_{A0} + A_p x_p}{\beta_e} \frac{dP_A}{dt} \\ Q_B = A_p \frac{dx_p}{dt} + C_{ip} P_L - C_{ep} P_B - \frac{V_{B0} - A_p x_p}{\beta_e} \frac{dP_B}{dt} \end{cases}, \quad (4)$$

where A_p is the effective actuating area of the cylinder, x_p is the cylinder displacement, C_{ip} is the internal leakage coefficient, C_{ep} is the external leakage coefficient, β_e is the oil effective bulk modulus, and V_{A0} and V_{B0} are initial left and right chamber volumes, respectively. As the piston of the actuator for the EHFS is initially centered, the initial volumes for the left and right chambers equal half of the total chamber volume V_t . Thus, Eq. (4) can be transformed to

$$Q_L = A_p \frac{dx_p}{dt} + C_{tp} P_L + \frac{V_t}{4\beta_e} \frac{dP_L}{dt}, \quad (5)$$

where $C_{tp} = C_{ip} + C_{ep}/2$ is the total leakage coefficient. Applying Newton's second law and ignoring the friction of the cylinder and the mass of oil, one can obtain the force F_g exerted on the payload

$$F_g = P_L A_p = m_t \frac{d^2 x_p}{dt^2} + B_p \frac{dx_p}{dt} + K x_p, \quad (6)$$

where m_t is the total mass with respect to the piston, B_p is the viscous damping coefficient, and K is the force-spring coefficient. The relationship between the control signal i_A and the spool position x_v of the servo valve can be modeled by [3]

$$G_{sv}(s) = \frac{X_v(s)}{I_A(s)} = \frac{K_{sv}}{\left(\frac{s^2}{\omega_{sv}^2} + \frac{2\xi_{sv}}{\omega_{sv}} s + 1 \right)}, \quad (7)$$

where K_{sv} , ω_{sv} , and ξ_{sv} are the gain, the angular natural frequency, and the damping coefficient of the servo valve, respectively.

By applying Laplace transform to Eqs. (1), (5), and (6), the open-loop force transfer function $G_{open}(s)$ from the servo valve spool displacement x_v to the exerted force F_g can be described by

$$G_{open}(s) = \frac{\frac{K_q}{A_p} (m_t s^2 + B_p s + K)}{\frac{V_t m_t}{4\beta_e A_p^2} s^3 + \left(\frac{K_{ce} m_t}{A_p^2} + \frac{V_t m_t}{4\beta_e A_p^2} \right) s^2 + \left(1 + \frac{K_{ce} B_p}{A_p^2} + \frac{V_t K}{4\beta_e A_p^2} \right) s + \frac{K_{ce} K}{A_p^2}}, \quad (8)$$

which can be further deduced to

$$G_{open}(s) = \frac{\frac{K_q A_p}{K_{ce}} \left(\frac{s^2}{\omega_m^2} + \frac{2\xi_m}{\omega_m} s + 1 \right)}{\left(\frac{s}{\omega_r} + 1 \right) \left(\frac{s^2}{\omega_0^2} + \frac{2\xi_0}{\omega_0} s + 1 \right)}, \quad (9)$$

where $\omega_m = \sqrt{K/m_t}$ and $\xi_m = B_p/(2\sqrt{K m_t})$ are the angular natural frequency and the damping ratio of the payload, respectively, $\omega_r = K_{ce} K/[A_p^2(1 + K/K_h)]$ is the first-order inertial element of the EHFS, $\omega_0 = \omega_h \sqrt{1 + K/K_h}$ and $\xi_0 = 2\beta_e K_{ce} K_h/[V_t(K_h + K)\omega_0] + B_p/(2m_t \omega_0)$ are the angular natural frequency and the damping ratio of the EHFS, respectively, $\omega_h = \sqrt{4\beta_e A_p^2/(m_t V_t)}$ is the angular natural frequency of the hydraulic cylinder, ξ_h is hydraulic damping ratio of actuator defined by $\xi_h = ((K_c + C_{tp})/A_p) \sqrt{\beta_e m_t/V_t} + (B_p/4A_p) \sqrt{V_t/\beta_e m_t}$, $K_h = m_t \omega_h^2$ is the oil spring stiffness, and $K_{ce} = K_c + C_{tp}$.

In [3], the magnitude and phase frequency characteristics of the force open-loop system from simulation and experiment are shown in Fig. 3 with some main parameters of the hydraulic dynamic model listed in Table 2. It can be seen that the experimental results are close to the simulation results, which demonstrates that the established hydraulic dynamic model is in high accordance with the actual EHFS.

3. Controller design

Fig. 4 presents an overall control scheme for the EHFS. As depicted in the figure, the proposed method includes a PI controller,

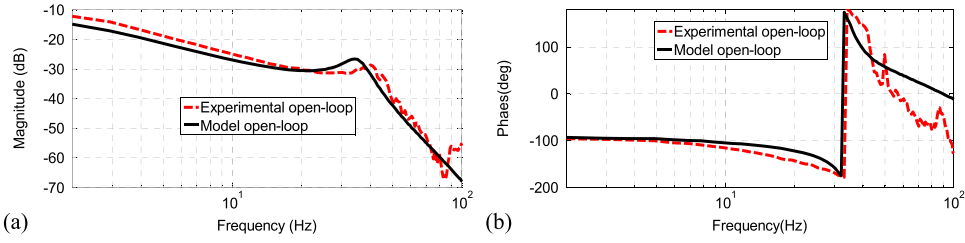


Fig. 3. Frequency response characteristics of the open-loop EHFS: (a) magnitude and (b) phase [3].

Table 2

Main parameters of the simulation model of the EHFS.

Parameters	Values
Effective actuating area of the cylinder, A_p	$1.88 \times 10^{-3} \text{ m}^2$
Viscous damping coefficient, B_p	25,000 Ns/m
External leakage coefficient, C_{ep}	$4.6 \times 10^{-17} \text{ m}^3/(\text{s}/\text{Pa})$
Internal leakage coefficient, C_{ip}	$4.6 \times 10^{-17} \text{ m}^3/(\text{s}/\text{Pa})$
Force-spring coefficient, K	$2 \times 10^7 \text{ N/m}$
Linearized flow gain coefficient, K_q	0.00145 ($\text{m}^3/\text{s}/\text{V}$)
Linearized flow pressure coefficient, K_c	$2 \times 10^{-12} \text{ m}^3/(\text{s}/\text{Pa})$
Gain of servo valve, K_{sv}	4 $\text{m}^3/\text{s}/\text{A}$
Total mass with respect to the piston, m_t	500 kg
Total chamber volume, V_t	$0.96 \times 10^{-3} \text{ m}^3$
Angular natural frequency of hydraulic cylinder, ω_h	32 Hz
Angular natural frequency of servo valve, ω_{sv}	100 Hz
Damping ratio of hydraulic cylinder, ξ_h	0.35
Damping ratio of servo valve, ξ_{sv}	0.7
Oil effective bulk modulus, β_e	$6.9 \times 10^8 \text{ Pa}$

an ODFC, and an online adaptive controller. In the first phase (Step 1 shown in Fig. 4), the PI controller is applied to the force closed-loop system of the EHFS, which guarantees force tracking accuracy. In the next stage (Step 2 shown in Fig. 4), the PI-controlled EHFS is excited by random force input signals and input-output data are acquired to realize identification of the PI-controlled force closed-loop system using a continuous system identification algorithm (CSIA). With the identified nominal model, an optimized feedback controller is designed by employing the H_∞ theory to further improve the force tracking performance. Finally (Step 3 shown in Fig. 4), an online adaptive controller on the basis of the ODFC is employed as an outer loop of the PI-controlled force closed-loop system to deal with the varying dynamics and nonlinearity of the EHFS so as to improve the force tracking accuracy. The functionalities for different components of the proposed controller can be summarized as follows:

(a) The PI controller is employed to stabilize the whole electro-

hydraulic force closed-loop system and obtain a preliminary force tracking performance, which lays a foundation for the subsequent controller design process.

- (b) The ODFC has the feasibility of extending the frequency bandwidth and alleviating the phase lag phenomenon for the force closed-loop system by the PI controller, which greatly enhances force tracking accuracy. Besides, the ODFC creates a great potential for the subsequent adaptive controller in the aspect of improving tracking performance.
- (c) The functionality for the outer loop online adaptive controller is to deal with defects of the previous offline designed controller in terms of system varying dynamics, unmodeled nonlinearities and external disturbances, which can further improve high-fidelity force tracking accuracy after the adaptive algorithm converges to its optimal solution.

3.1. PI controller

A classical PI controller is employed to constitute the force closed-loop control system of the EHFS and its block diagram is demonstrated in Fig. 5, where $r_F(t)$ is the reference force signal, $y_F(t)$ is the force feedback signal measured from the EHFS. As the bandwidth of the servo valve is considerably larger than that of the EHFS, and therefore the dynamics of the servo valve is not included in Fig. 5, so as to simplify the control design process. The transfer function $G_{PI}(s)$ of the PI controller of the force closed-loop system is expressed by

$$G_{PI}(s) = \frac{K_P s + K_I}{s}, \quad (10)$$

where K_P and K_I represent the proportional and integral coefficients, respectively.

By simplifying the block diagram of Fig. 5, the force closed-loop transfer function $G_F(s)$ of the EHFS with combination of the PI controller shown in Eq. (10) and the force open-loop system shown in Eq. (9) can be deduced by

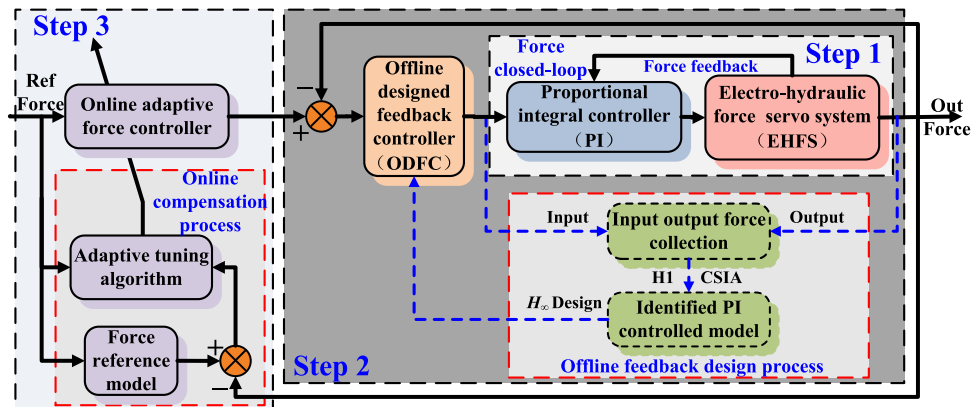


Fig. 4. Proposed controller for the EHFS.

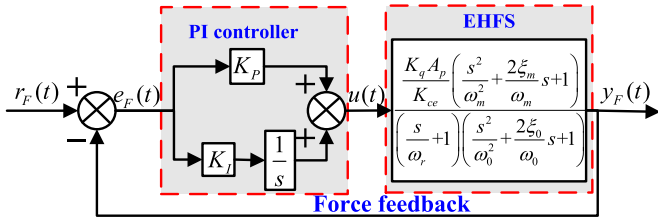


Fig. 5. Force closed-loop system with the PI controller for the EHFS.

$$G_F(s) = \frac{G_{open}(s)G_{PI}(s)}{1 + G_{open}(s)G_{PI}(s)} = \frac{b_0 s^3 + b_1 s^2 + b_2 s + b_3}{s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4}, \quad (11)$$

where the polynomial coefficients are given by

$$\begin{cases} b_0 = K_q A_p K_P \omega_r \omega_0^2 / (K_{ce} \omega_m^2) \\ b_1 = [2K_q A_p K_P \xi_m \omega_r \omega_0^2 \omega_m + K_q A_p K_I \omega_r \omega_0^2] / K_{ce} \omega_m^2 \\ b_2 = [K_q A_p K_P \omega_r \omega_0^2 \omega_m + 2K_q A_p \xi_m K_I \omega_r \omega_0^2] / K_{ce} \omega_m \\ b_3 = K_q A_p K_I \omega_r \omega_0^2 / K_{ce} \end{cases} \quad (12)$$

$$\begin{cases} a_1 = 2\xi_0 \omega_0 + \omega_r + K_q A_p K_P \omega_r \omega_0^2 / K_{ce} \omega_m^2 \\ a_2 = \omega_0^2 + 2\xi_0 \omega_r + [2K_q A_p K_P \xi_m \omega_r \omega_0^2 \omega_m + K_q A_p K_I \omega_r \omega_0^2] / K_{ce} \omega_m^2 \\ a_3 = \omega_r \omega_0^2 + [K_q A_p K_P \omega_r \omega_0^2 \omega_m + 2K_q A_p \xi_m K_I \omega_r \omega_0^2] / K_{ce} \omega_m \\ a_4 = K_q A_p K_I \omega_r \omega_0^2 / K_{ce} \end{cases} \quad (13)$$

3.2. Continuous system identification algorithm

As the nominal EHFS model controlled by the PI controller is required in the design process of the ODFC, the PI-controlled EHFS must be firstly established so as to provide the plant information. The force closed-loop transfer function of the PI-controlled EHFS is given in Eq. (11), but it is difficult to determine the polynomial coefficients of the system model since it is difficult to accurately measure many hydraulic parameters of the EHFS. To resolve this problem, a CSIA is employed to identify the polynomial coefficients in Eq. (11). To implement the CSIA, the force frequency response sequence $\{\omega_i, G_{FC}(\omega_i)\}_{i=1,2,\dots,N}$ of the PI-controlled EHFS must be obtained in advance. With the experimentally acquired input and output force data, the force frequency response sequence of the EHFS can be readily calculated using the H1 estimation approach [30], whose basic principle can be described by

$$G_{FC}(\omega) = \frac{P_{yx}(\omega)}{P_{xx}(\omega)}, \quad (14)$$

where $P_{yx}(\omega)$ represents the cross-power spectral density of the input and output force signals, and $P_{xx}(\omega)$ is the auto-power spectral density of the reference input force signal. The basic principle of the CSIA is to find a suitable polynomial coefficient so that for any prescribed frequency response sequence $\{\omega_k, G_{FC}(\omega_k)\}$, the calculated frequency response according to Eq. (11) equals the experimentally acquired result. Based on this principle, substituting the definition $s = j\omega_k$ into Eq. (11) yields

$$G_{FC}(\omega_k) = \frac{b_0(j\omega_k)^3 + b_1(j\omega_k)^2 + b_2(j\omega_k) + b_3}{(j\omega_k)^4 + a_1(j\omega_k)^3 + a_2(j\omega_k)^2 + a_3(j\omega_k) + a_4} \quad (k = 1, 2, \dots, N). \quad (15)$$

Substituting $G_{FC}(j\omega_k) = R(\omega_k) + jI(\omega_k)$ into Eq. (15) and separating both sides of the simplified equation by real and imaginary components yield

$$\begin{aligned} R(\omega_k) + jI(\omega_k) &= \left[-\frac{1}{\omega_k^2} b_1 + \frac{1}{\omega_k^4} b_3 - \frac{I(\omega_k)}{\omega_k} a_1 + \frac{R(\omega_k)}{\omega_k^2} a_2 + \frac{I(\omega_k)}{\omega_k^3} a_3 - \frac{R(\omega_k)}{\omega_k^4} a_4 \right] \\ &+ j \left[-\frac{1}{\omega_k} b_0 + \frac{1}{\omega_k^3} b_2 + \frac{R(\omega_k)}{\omega_k} a_1 + \frac{I(\omega_k)}{\omega_k^2} a_2 - \frac{R(\omega_k)}{\omega_k^3} a_3 - \frac{I(\omega_k)}{\omega_k^4} a_4 \right], \end{aligned} \quad (16)$$

where $R(\omega_k)$ and $I(\omega_k)$ are real and imaginary components of experimentally calculated $G_{FC}(\omega_k)$, respectively. Equating the real and imaginary parts of both sides of Eq. (16) yields

$$\begin{bmatrix} \mathbf{h}_r(\omega_k) \\ \mathbf{h}_i(\omega_k) \end{bmatrix} \boldsymbol{\theta} = \begin{bmatrix} R(\omega_k) \\ I(\omega_k) \end{bmatrix} \quad (k = 1, 2, \dots, N), \quad (17)$$

where $\mathbf{h}_r(\omega_k)$, $\mathbf{h}_i(\omega_k)$, and $\boldsymbol{\theta}$ are vectors defined by

$$\begin{cases} \mathbf{h}_r(\omega_k) = \left[0, -\frac{1}{\omega_k^2}, 0, \frac{1}{\omega_k^4}, -\frac{I(\omega_k)}{\omega_k}, \frac{R(\omega_k)}{\omega_k^2}, \frac{I(\omega_k)}{\omega_k^3}, -\frac{R(\omega_k)}{\omega_k^4} \right] \\ \mathbf{h}_i(\omega_k) = \left[-\frac{1}{\omega_k}, 0, \frac{1}{\omega_k^3}, 0, \frac{R(\omega_k)}{\omega_k}, \frac{I(\omega_k)}{\omega_k^2}, -\frac{R(\omega_k)}{\omega_k^3}, \frac{I(\omega_k)}{\omega_k^4} \right] \\ \boldsymbol{\theta} = [b_0, b_1, b_2, b_3, a_1, a_2, a_3, a_4]^T \end{cases} \quad (18)$$

Combining N sets of frequency response sequences in Eq. (18) and rewriting them in one formulation yield

$$\mathbf{H}_{2N \times (m+n+1)} \boldsymbol{\theta}_{(m+n+1) \times 1} = \boldsymbol{\Psi}_{2N \times 1}, \quad (19)$$

where $\mathbf{H}$ and $\boldsymbol{\Psi}$ are defined by

$$\begin{cases} \mathbf{H} = [\mathbf{h}_r^T(\omega_1) \ \mathbf{h}_i^T(\omega_1) \ \mathbf{h}_r^T(\omega_2) \ \mathbf{h}_i^T(\omega_2) \ \dots \ \mathbf{h}_r^T(\omega_N) \ \mathbf{h}_i^T(\omega_N)]^T \\ \boldsymbol{\Psi} = [R(\omega_1) \ I(\omega_1) \ R(\omega_2) \ I(\omega_2) \ \dots \ R(\omega_N) \ I(\omega_N)]^T \end{cases} \quad (20)$$

There generally exist many more equations in Eq. (19) than the number of polynomial coefficients to be determined; hence accurate solutions of Eq. (19) cannot be obtained. In fact, only the approximate solution for the parameter vector $\boldsymbol{\theta}$ can be obtained using a least-square method and the approximate solution is described by

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \boldsymbol{\Psi}. \quad (21)$$

In actual implementation, the matrix $\mathbf{H}^T \mathbf{H}$ is likely to be ill-conditioned since the matrix dimension is huge; the calculated parameter vector $\boldsymbol{\theta}$ might not accurately reflect the system's dynamics. To cope with this problem, the coefficient vector $\boldsymbol{\theta}$ deduced from Eq. (21) is regarded as an initial value for the following optimization problem

$$\arg \min_{\boldsymbol{\theta}} \sum_{k=1}^N W(\omega_k) \left| G(\omega_k) - \frac{N(\omega_k)}{D(\omega_k)} \right|^2, \quad (22)$$

where $W(\omega_k)$ is the weighting function and can be selected as the inverse of the frequency response data so as to minimize a relative error rather than an absolute error. In order to get the optimized parameter vector that minimizes the cost function, a damped Gauss-Newton iterative method [31] is further employed.

3.3. Offline designed feedback controller

The real-time feedback controller for the EHFS is offline designed by a H_∞ robust control method, which takes the system performance, control efforts, and system robustness into consideration at the same time to obtain the optimized controllers. The block diagram of the real-time feedback system for robust analysis is shown in Fig. 6, where $r_F(k)$, $e_F(k)$, and $y_F(k)$ are the reference force input, the force tracking error, and the actual force output signal, respectively, $u(k)$ represents the feedback control input, W_p , W_U , and W_T are weighting functions imposing

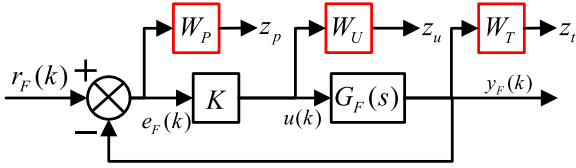


Fig. 6. Block diagram of the real-time feedback system for H_∞ analysis.

requirements on the EHFS tracking performance, the control input magnitude, and robustness, respectively, and z_p , z_u , and z_t are the weighted outputs.

By simplifying Fig. 6, the formulations from the reference force input $r_F(k)$ to the tracking error $e_F(k)$, the force output $y_F(k)$, and the feedback control input $u(k)$, can be described by

$$\begin{cases} e_F(k) = \frac{1}{1 + KG_F(s)} r_F(k) = S r_F(k) \\ y_F(k) = \frac{KG_F(s)}{1 + KG_F(s)} r_F(k) = T r_F(k), \\ u(k) = \frac{K}{1 + KG_F(s)} r_F(k) = R r_F(k) \end{cases} \quad (23)$$

where S is the sensitivity function, T is the complementary sensitivity function, and R is the feedback control input sensitivity function. With the desired weighting functions, the block diagram of the feedback EHFS can be transformed in the standard H_∞ configuration as depicted in Fig. 7, and the following equation can be obtained:

$$\begin{bmatrix} z \\ e_F \end{bmatrix} = P_n(s) \begin{bmatrix} v \\ u \end{bmatrix} = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \begin{bmatrix} v \\ u \end{bmatrix} = \begin{bmatrix} W_P & -W_P G_F(s) \\ 0 & W_U \\ 0 & W_T G_F(s) \\ 1 & -G_F(s) \end{bmatrix} \begin{bmatrix} v \\ u \end{bmatrix}, \quad (24)$$

where $P_n(s)$ is the generalized plant, z is the weighted outputs defined by $z = [z_p, z_u, z_t]^T$, and v is the exogenous input signal. The transfer function $T_{zv}(s)$ from the external input signals v to the weighted outputs z can be deduced by

$$\begin{aligned} T_{zv}(s) &= F_l(P_n(s), K) = P_{11}(s) + P_{12}(s)K(I - P_{22}(s)K)^{-1}P_{21}(s) \\ &= \begin{bmatrix} W_P S \\ W_U R \\ W_T T \end{bmatrix}, \end{aligned} \quad (25)$$

where $F_l(P_n(s), K)$ is the lower linear fractional transformation that is usually employed to indicate the incorporation of a controller K into a system. Therefore, the following transfer function matrix inequality is obtained for the optimized feedback controller:

$$\|T_{zv}(s)\|_\infty = \sup_{\omega} \sigma_{\max}(T_{zv}(j\omega)) \leq \gamma_{\min} < \gamma, \quad (26)$$

where $\sigma_{\max}(T_{zv}(j\omega))$ is the largest singular value of $T_{zv}(j\omega)$.

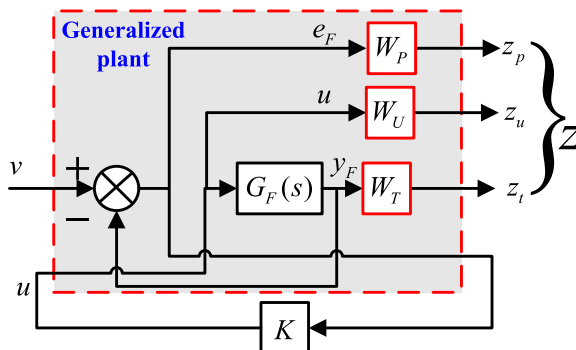


Fig. 7. Standard configuration for the ODFC.

Once all the three weighting functions W_P , W_U , and W_T are chosen, the feedback controller K can be readily obtained by solving two Riccati formulae with the γ -iteration technique to determine the optimal value of γ . However, the selection of weighting functions for a specified problem often involves many iterations, and fine tuning, and it is very hard to give a general formula for the weighting functions that would work in every case. Nevertheless, there are still some universal criteria for choosing these weighting functions. The performance weighting function W_P is imposed on the sensitivity function of the EHFS, and in order to obtain a small tracking error, the magnitude of the sensitivity function should be kept small over the interested frequency bandwidth, which indicates that the performance weighting function should exhibit high-gain properties within the interested frequency. Therefore, the performance weighting function W_P can be chosen as a low-pass system with high gains at low frequencies. The control input weighting function W_U is employed to limit the magnitude of the control input signal and to prevent severe saturation so that the EHFS actuates within a tolerant range. Besides, with the application of the control input weighting function, the singular problem that occurs when solving two Riccati formulae can be readily avoided. Hence, a small constant value is usually adopted for the control input weighting function. The complementary sensitivity weighting function W_T is used to bound multiplicative uncertainties of the EHFS, and therefore the magnitude of W_T should be chosen to cover that of the system's modeling error, i.e.,

$$|W_T| > \left| \frac{G_F(s)}{\hat{G}_F(s)} - 1 \right|. \quad (27)$$

Due to the fact that the modeling error is enlarged with the increase of the frequency, to ensure the system's robustness when uncertainties exist, the complementary sensitivity weighting function is usually selected as a high-pass transfer function with high gains at high frequencies.

3.4. Online adaptive compensator

To further enhance the force tracking performance of the EHFS, an online adaptive controller is cascaded to the EHFS compensated by the ODFC, and the block diagram is presented in Fig. 8, where the adaptive filter is online tuned by the nLMS algorithm using the reference force input and the tracking error. The detailed structure configuration of the adaptive filter is demonstrated in Fig. 9, where the adaptive controller consists of a tapped delay line, adjustable weights with the tapped delay line output as the input, a sum-mator to add the weighted outputs, and an adaptive mechanism. The output signal of the adaptive controller to the robust controlled EHFS can be described by

$$u(k) = \mathbf{r}_F(k)^T \mathbf{w}(k) = \sum_{i=0}^{m-1} r_F(k-i) w_i(k), \quad (28)$$

where k is the k th sample time, $\mathbf{r}_F(k)$ is the reference force input vector defined by $\mathbf{r}_F(k) = [r_F(k), r_F(k-1), \dots, r_F(k-m+1)]^T$, and $\mathbf{w}(k)$ is the online adaptive weight defined by $\mathbf{w}(k) = [w_0(k), w_1(k), \dots, w_{m-1}(k)]^T$.

The desired system force output signal for the adaptive controller in Fig. 8 is generated from a pure time delay process z^{-L} , and the tracking error $e_F(k)$ in the z -domain $E_F(z)$ can be written as

$$\begin{aligned} E_F(z) &= z^{-L} R_F(z) - U(z) \frac{KG_F(z)}{1 + KG_F(z)} \\ &= z^{-L} R_F(z) - R_F(z) G_W(z) \frac{KG_F(z)}{1 + KG_F(z)}, \end{aligned} \quad (29)$$

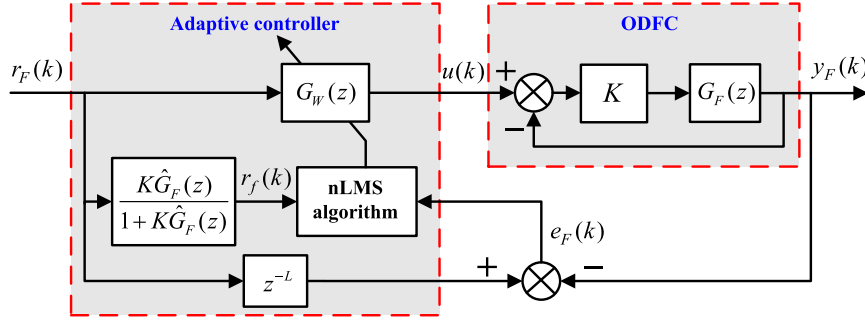


Fig. 8. Proposed control scheme for the EHFS.

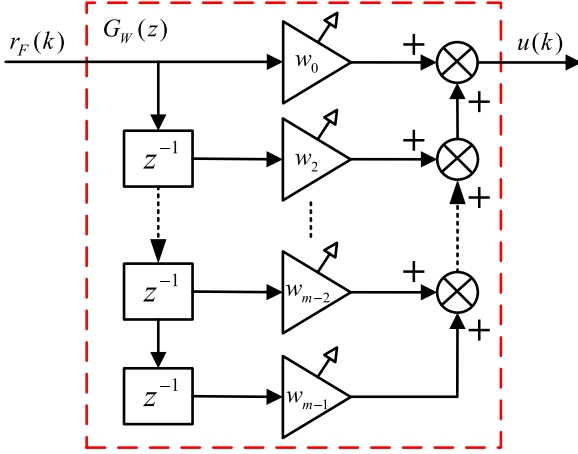


Fig. 9. Block diagram of the adaptive filter configuration.

where $G_W(z)$ is the transfer function of the adaptive filter. By employing Eq. (29), the derivative of $E_F(z)$ with respect to $G_W(z)$ is expressed by

$$\frac{dE_F(z)}{dG_W(z)} = -\frac{KG_F(z)}{1 + KG_F(z)} R_F(z), \quad (30)$$

and therefore the derivative of $e_F(k)$ with respect to $\mathbf{w}(k)$ can be expressed by

$$\frac{\partial e_F(k)}{\partial \mathbf{w}(k)} = -\left[\frac{KG_F(z)}{1 + KG_F(z)} \right] * \mathbf{r}_F(k), \quad (31)$$

where $*$ is the time domain convolution operation. With the cost function defined by $J(k) = e_F^2(k)$ and use of Eq. (31), the instantaneous measurement of the local gradient, which is the derivative of the squared error $e_F(k)$, can be simplified as

$$\begin{aligned} \frac{\partial J(k)}{\partial \mathbf{w}(k)} &= \frac{\partial e_F^2(k)}{\partial \mathbf{w}(k)} = 2e_F(k) \frac{\partial e_F(k)}{\partial \mathbf{w}(k)} \\ &= -2 \left[\frac{KG_F(z)}{1 + KG_F(z)} \right] * \mathbf{r}_F(k) e_F(k). \end{aligned} \quad (32)$$

As can be seen from Eq. (32), the input force $r_F(k)$ should be firstly filtered by the H_∞ feedback controlled EHFS model before multiplication of the force tracking error to form the instantaneous gradient. However, the actual EHFS model controlled by the ODFC is not available in real applications, and to overcome this problem, the estimated force closed-loop system model is employed, where the filtered input signal force $r_f(k)$ can be described by

$$\mathbf{r}_f(k) = \left[\frac{K\hat{G}_F(z)}{1 + K\hat{G}_F(z)} \right] * \mathbf{r}_F(k). \quad (33)$$

and the nLMS algorithm for updating the weights in the negative

instantaneous gradient is depicted in Fig. 10 and it can be written as

$$\mathbf{w}_{i+1}(k) = \mathbf{w}_i(k) + \lambda e_F(k) \frac{\mathbf{r}_f(k)}{\|\mathbf{r}_f(k)\|}, \quad (34)$$

where λ is the iterative step size, and $\|\mathbf{r}_f(k)\|$ is the norm of the filtered force input vector.

To ensure convergence of the adaptive algorithm, the phase error of the estimated H_∞ controlled EHFS model should be less than 90° compared with the actual plant model [2,32], i.e.

$$\angle \left(\frac{K\hat{G}_F(z)}{1 + K\hat{G}_F(z)} \right) - \angle \left(\frac{KG_F(z)}{1 + KG_F(z)} \right) < 90^\circ, \quad (35)$$

and this condition can be easily satisfied as long as the error between $\hat{G}_F(z)$ and $G_F(z)$ is sufficiently small.

4. Experimental studies

4.1. Experimental control system

The experimental control system scheme is briefly depicted in Fig. 11, where the xPC rapid prototype technology comprised a host computer and a target computer is utilized for force control purpose. An Advantech industrial computer is served as the target computer, and it is installed with the xPC target real-time operating system (RTOS), an A/D data acquisition card, and a D/A signal driven card. The host computer is a personal Windows XP laptop with the MATLAB/Simulink and Microsoft Visual Studio software. A cross twisted-pair cable is employed for communication between the host computer and the target computer by TCP/IP protocol.

Fig. 12 displays the experimental MATLAB/Simulink model with different controllers applied to the EHFS, including the Force Generator, the Normal Motion Logic, the Gcd based on the ODFC, the adaptive LMS algorithm (adp_LMS), and A/D card and D/A card driver programs (PCI-1716 and ACL-6126). The feedback force signal is firstly converted and amplified by the conditioning module from a current in the range of 4–20 mA to a voltage in the range of 2–10 V, and it is then collected by the 16-bit Advantech A/D card PCI-1716 on the target computer. The collected voltage is calibrated by the conditioning module from the range of 2–10 V to

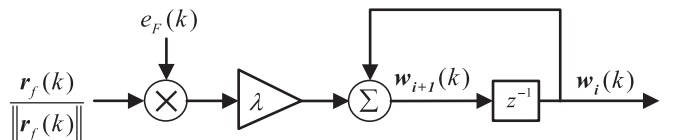


Fig. 10. Block diagram of the LMS adaptive algorithm for weight updating.

the range of -10 V to 10 V , which can be seen from Figs. 11 and 12. The servo valve drive signal is sent by the target computer through a 12-bit Adlink D/A card ACL-6126 with the voltage in the range of -10 V to 10 V before it is transformed to a current in the range of -40 mA to 40 mA for the servo valve by the conditioning module, which can be seen from Figs. 11 and 12. The different controllers are programmed and compiled by MATLAB/Simulink and Visual C on the host computer, respectively, and then is downloaded to the target computer for real-time execution. The sampling rate for the test rig is selected as 1000 Hz in all experiments.

4.2. Experimental results of the PI controller

A force step response with 500 N is applied as a reference force input to tune the PI controller. Firstly, the proportional coefficient K_p is gradually increased until the force step output response just has an overshoot. Fig. 13(a) shows the proportional coefficient adjusting process from 1 s to 3 s and K_p is adjusted to be 0.0004 . After fine-tuning K_p , the integral coefficient K_i is gradually increased from 0 to reduce the stable force tracking error, until the force step response output just has an overshoot, as shown in Fig. 13(a) from 3 s to 4.9 s , and $K_i = 0.00002$. The PI controller with $K_p = 0.0004$ and

$K_i = 0.00002$, shown in Fig. 13(b), has ensured the force closed-loop to have a better force tracking performance with enough stability.

4.3. Experimental results of the force closed-loop system identification

After the force closed-loop system is well tuned by the PI controller, the next step is to employ the CSIA for system identification so as to provide the nominal model for offline design of the feedback controller. To effectively reflect the system's dynamics, a random force signal with an amplitude of 5000 N and a frequency of $0\text{--}50\text{ Hz}$ is input to the PI-controlled EHFS, and the real-time collected reference input and experimental output force signals are depicted in Fig. 14. With reference to the CSIA discussed in Section 3.2, the final identified continuous transfer function of the PI-controlled EHFS is expressed by

$$G_f(s) = \frac{26.11s^3 + 2.015e4s^2 + 2.337e6s + 1.505e8}{s^4 + 244.1s^3 + 4.513e4s^2 + 3.149e6s + 1.482e8}. \quad (36)$$

To evaluate the identified model precision by the CSIA, the frequency response of the identified model and the reference

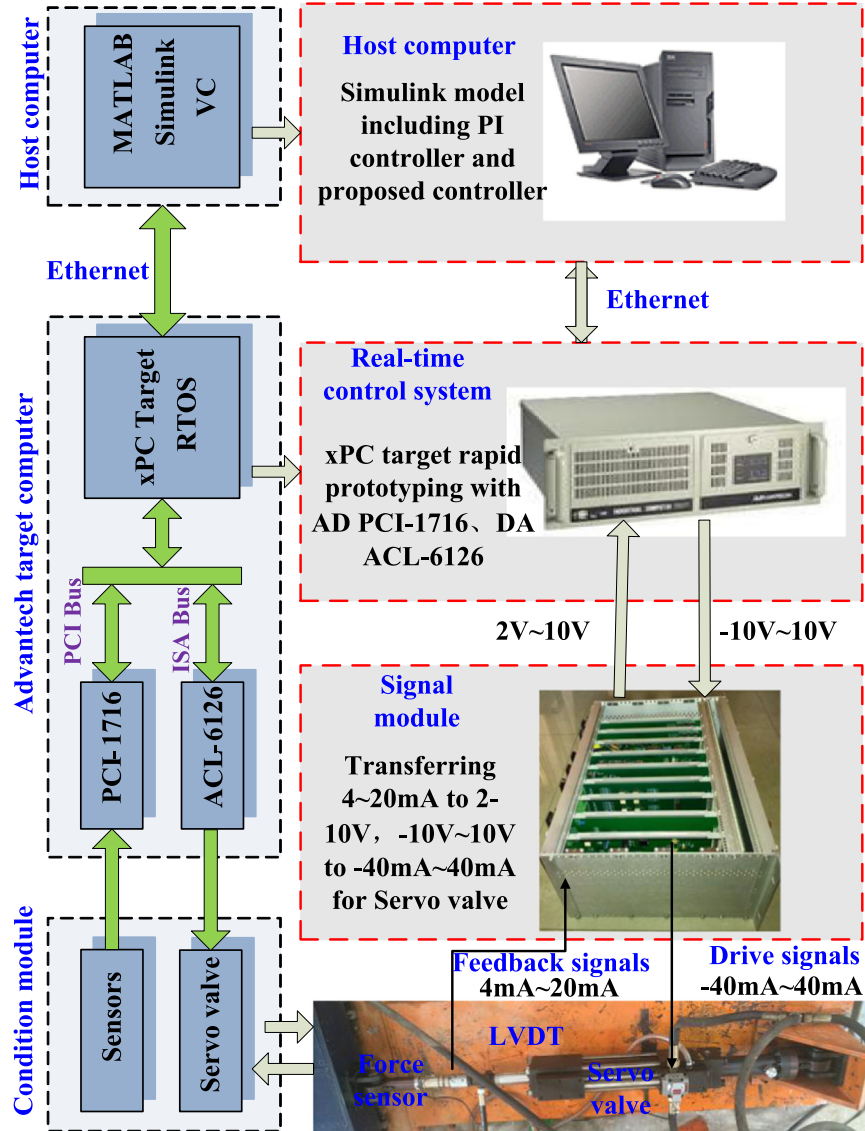


Fig. 11. Block schematic of the EHFS control system.

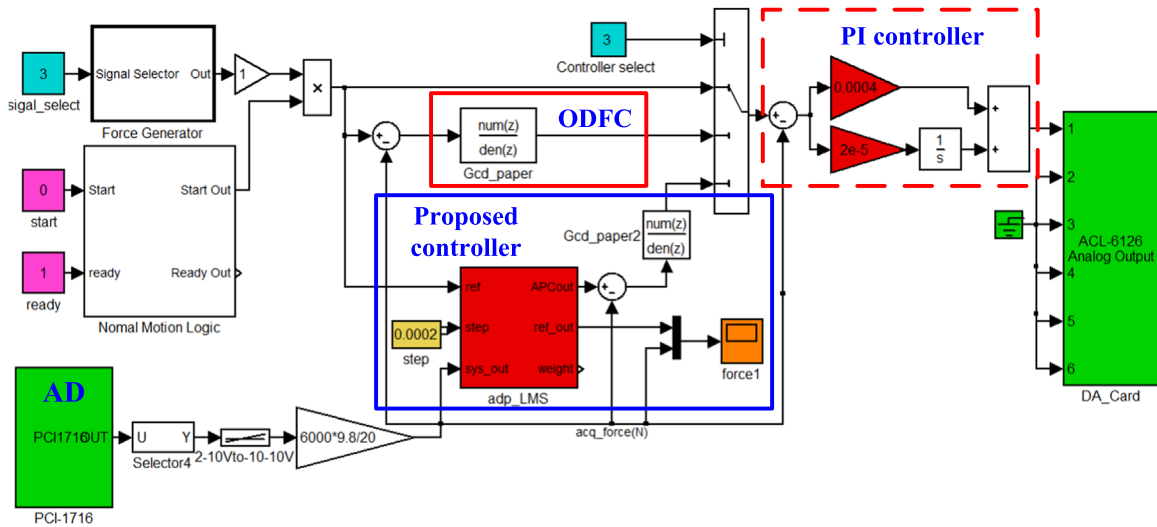


Fig. 12. Experimental MATLAB/Simulink model with different controllers for the EHFS.

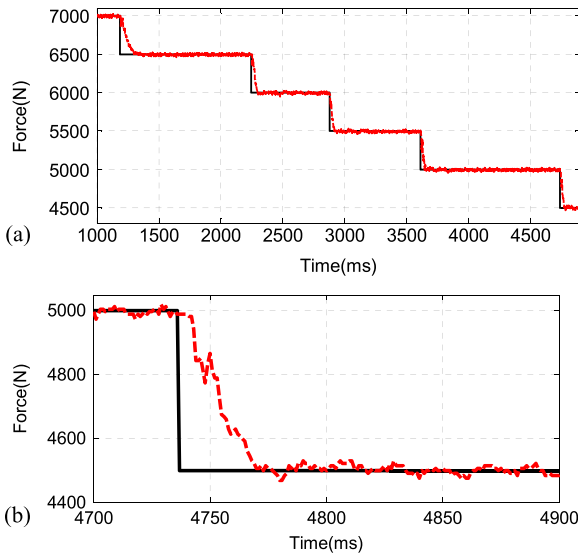


Fig. 13. Experimental results of the force step response with the PI controller: (a) the tuning process and (b) the tuned result.

experimental system are presented in Fig. 15, where the reference experimental model is achieved by the H1 estimation method. As can be seen from the figure, the identified model using the CSIA is in accordance with the experimental results, and identification accuracy is degraded as the frequency increases. Hence, the identified system model can well reflect the dynamic performance of the EHFS in the interested, relatively low frequencies.

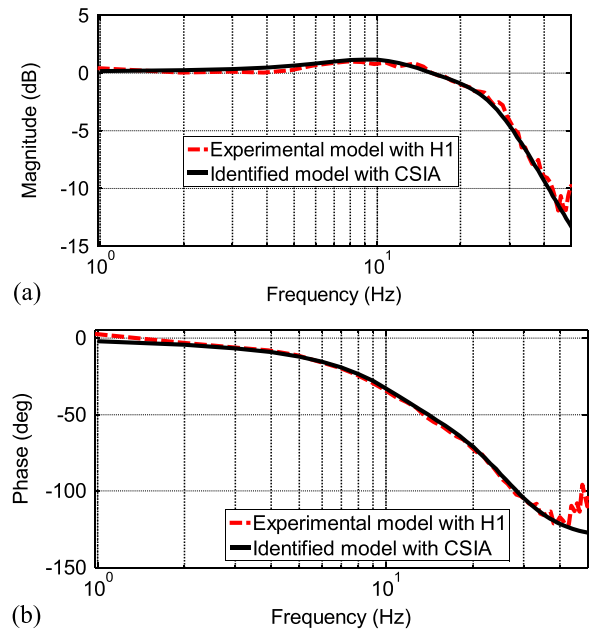


Fig. 15. Frequency responses characteristics of the experimental and identified EHFS with the H1 estimation method and the CSIA: (a) magnitude and (b) phase.

4.4. Experimental results of the ODFC

After the nominal model of the ODFC is obtained, the key issue left in the design process is to carefully select three weighting functions. Since the order of the final ODFC increases with those of

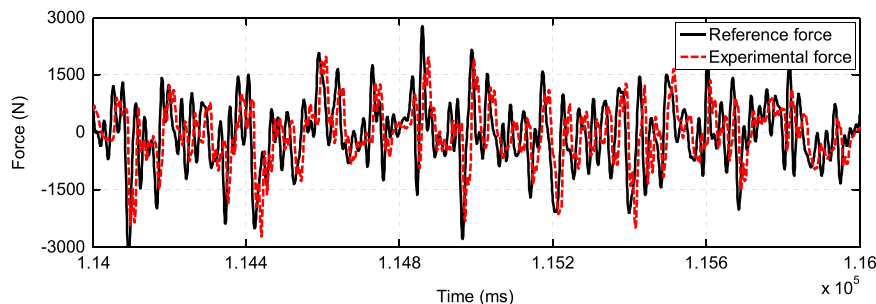


Fig. 14. Experimental results of the random force tracking waveform with the PI controller.

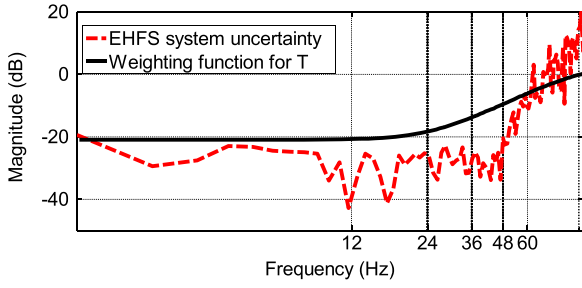


Fig. 16. Magnitude characteristics of the EHFS system uncertainty and the weighting function T .

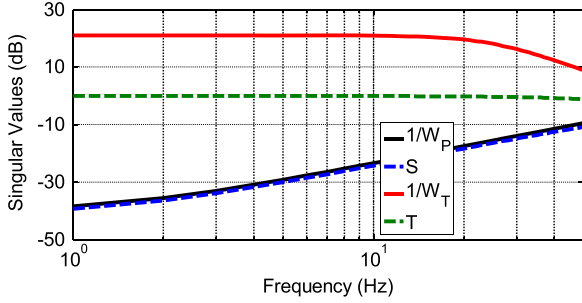


Fig. 17. Singular values of the inverse weighting, sensitivity and complementary sensitivity functions.

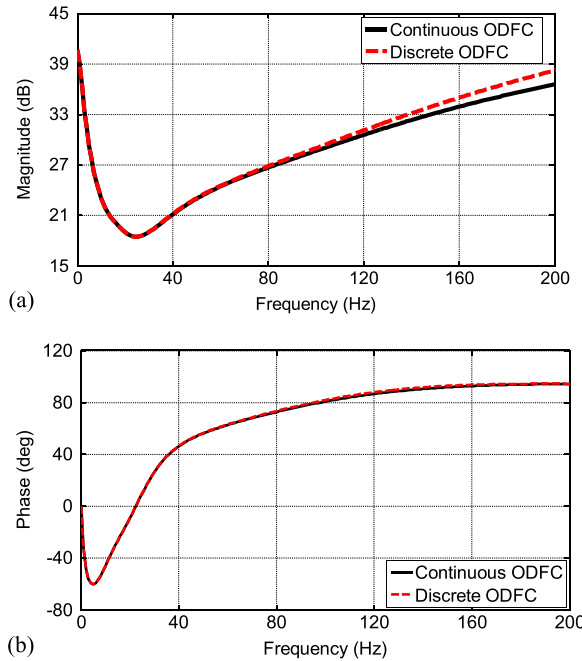


Fig. 18. Frequency response characteristics with the continuous and discrete ODFCs: (a) magnitude and (b) phase.

the weighting functions, to achieve a relatively low-order controller, the performance weighting function W_P is chosen as a first-order model

$$W_P = \frac{0.01s + 9.425e4}{100s + 942.5}. \quad (37)$$

Due to the fact that the complementary sensitivity weighting function is selected to bound the EHFS' multiplicative uncertainties with respect to the estimated nominal model, the system's multiplicative error is firstly obtained by experiment on the basis of the nominal model, and then a second-order transfer

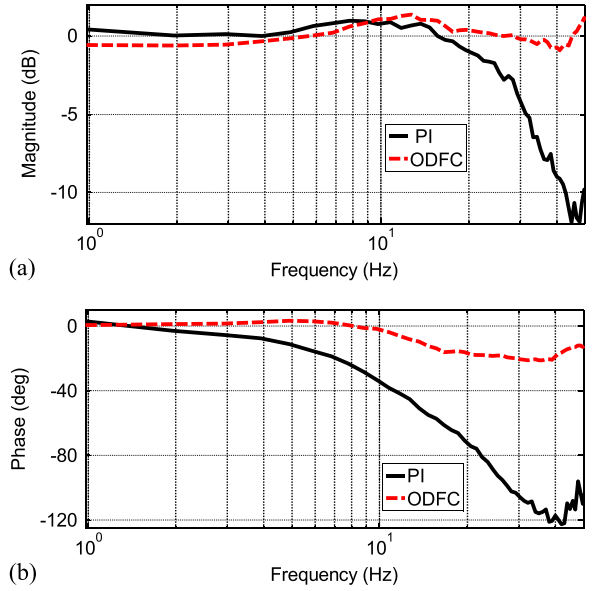


Fig. 19. Experimental results of frequency response characteristics using random force input signals with the PI and the ODFC: (a) magnitude and (b) phase.

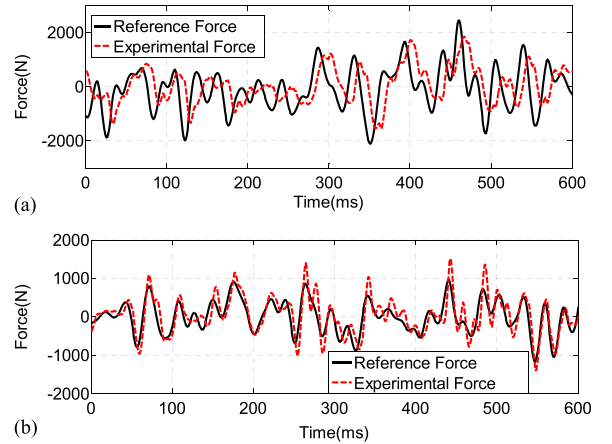


Fig. 20. Experimental results of random force input signals with (a) the PI controller and (b) the ODFC.

function is selected for the complementary sensitivity weight

$$W_T = \frac{3.612e - 6s^2 + 0.0007943s + 0.08913}{2.533e - 6s^2 + 0.002228s + 1}. \quad (38)$$

To verify the correctness of the selected complementary sensitivity weight, the magnitude of the multiplicative uncertainty and the complementary sensitivity weight are shown in Fig. 16. It can be obviously found that the employed weight satisfies the desired requirements. As for the control input weighting function, $W_U = 10^{-5}$ is used to simplify the design process. After the definition of the three weighting functions, the feedback robust controller is obtained by solving two Riccati formulae using the γ -iteration technique, and the final optimized controller is achieved by

$$K(s) = \frac{\left\{ \begin{aligned} &6.408e9s^6 + 7.201e12s^5 + 4.195e15s^4 + 8.922e17s^3 \\ &+ 1.329e20s^2 + 8.801e21s + 3.749e23 \end{aligned} \right\}}{\left\{ \begin{aligned} &s^7 + 3.795e6s^6 + 1.008e11s^5 + 1.408e14s^4 + 5.953e16s^3 \\ &+ 6.848e18s^2 + 4.286e20s + 3.48e21 \end{aligned} \right\}}. \quad (39)$$

Fig. 17 presents the singular values of the inverse weighting functions, and the sensitivity and complementary sensitivity

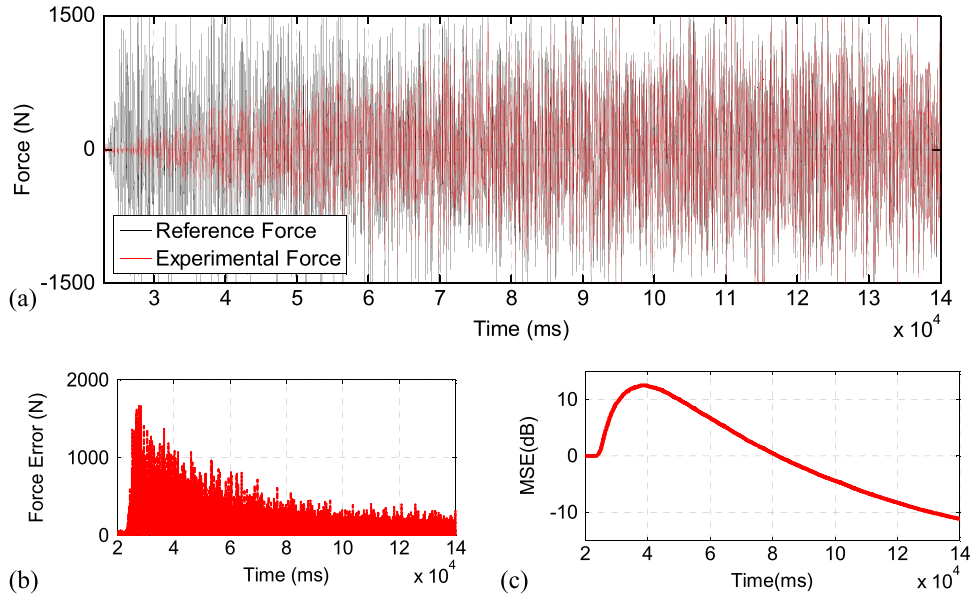


Fig. 21. Experimental results of the random force input signal using the proposed controller: (a) adaptive tuning process, (b) its tracking error, and (c) its MSE.

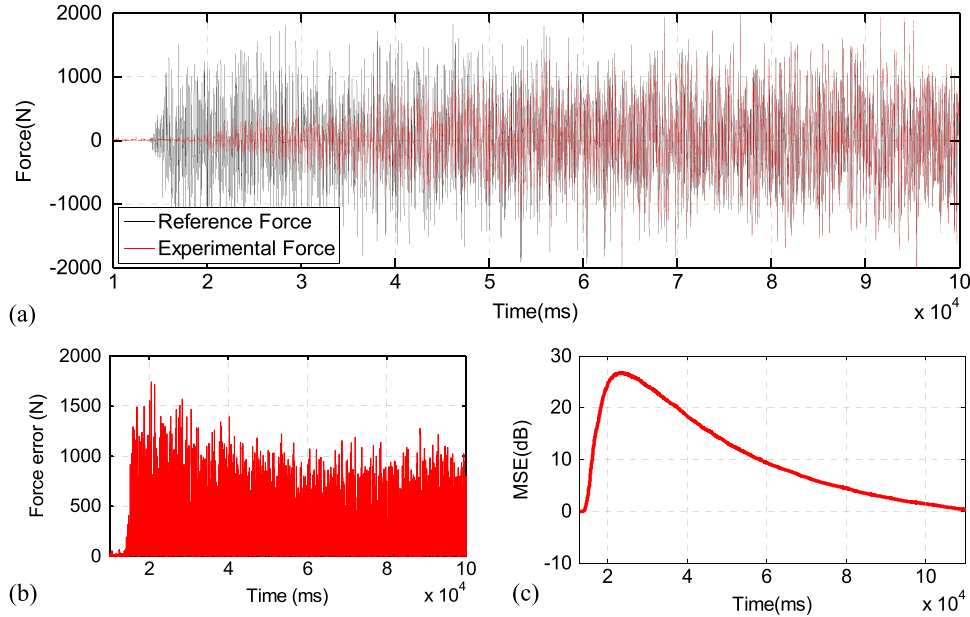


Fig. 22. Experimental results of the random force input using the direct adaptive controller: (a) adaptive tuning process and (b) its tracking error, and (c) its MSE.

functions for the analysis of the system's stability and performance. As can be seen from the figure, the singular values of the sensitivity function S and the complementary sensitivity function T lie below the inverse performance weight W_p and the inverse complementary sensitivity function W_t lie over the whole frequency bandwidth of interest, respectively, which indicates that the EHFS is stable in the presence of modeling uncertainties and performance specifications are met. In fact, this is in accordance with the fact that the calculated value γ should be less than unity to guarantee the global stability for the range of uncertainties, since the final optimized γ is obtained as $\gamma = 0.9103$. As shown in Eq. (39), the coefficients of the obtained continuous feedback controller are extremely huge, which results in large computation burden for the control routine. To simplify the implementation process, Tustin transform is employed in Eq. (39) so as to convert the continuous controller to the discrete form and the final obtained discrete controller is given by

$$K(z) = \frac{114.9z^7 - 449.8z^6 + 594.2z^5 - 115.8z^4 - 479.1z^3 + 527z^2 - 230z + 38.58}{z^7 - 2.021z^6 - 0.4401z^5 + 3.229z^4 - 1.492z^3 - 1.02z^2 + 0.9331z - 0.1893} \quad (40)$$

The frequency characteristics of the continuous and discrete ODFCs are shown in Fig. 18, from which it can be inferred that the discrete precision by Tustin method gradually deteriorates with the increase of the frequency. However, in the working frequencies below 50 Hz for the EHFS, the discrete controller $K(z)$ well matches the continuous controller $K(s)$. Therefore, the effectiveness of the discrete controller is proved. With the control program shown in Fig. 12, the experimental frequency responses of the ODFC controller together with the conventional PI controller is presented in Fig. 19, and the time history random force waveforms for the two controllers are also given in Fig. 20. It can be seen from Fig. 19(a) that the magnitude frequency bandwidth (within ± 3 dB) of the force closed-loop system are approximately 28 Hz and 50 Hz with the PI controller and the ODFC, respectively. Besides, Fig. 19(b) reveals that the phase delay has also been greatly

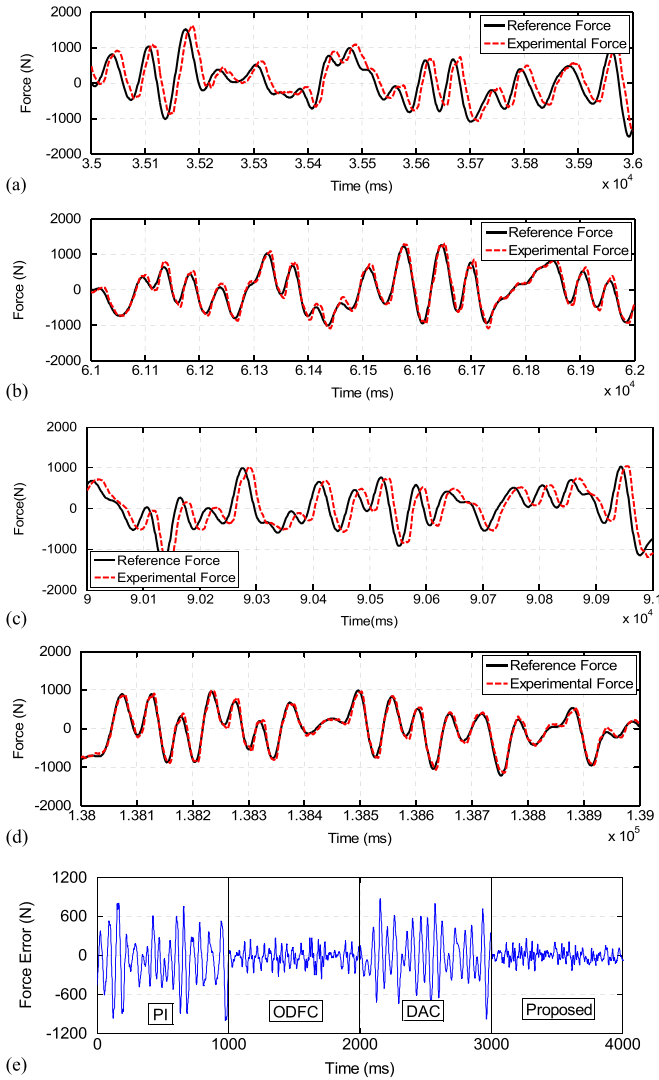


Fig. 23. Experimental results of the random force tracking performance with (a) the PI controller, (b) the ODFC, (c) the DAC, and (d) the proposed controller, and (e) their tracking errors.

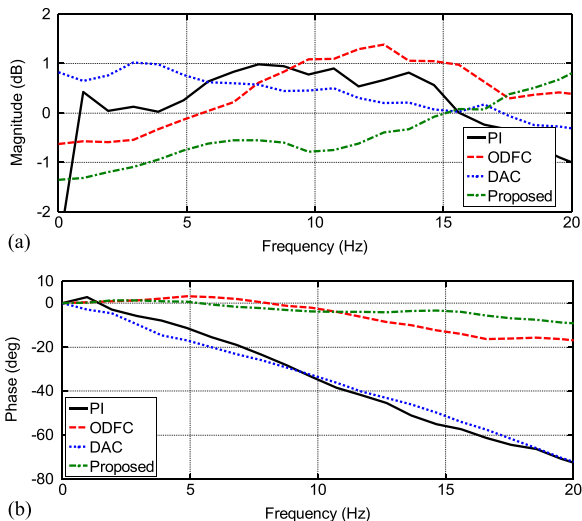


Fig. 24. Experimental results of frequency response characteristics using random force signals with different controllers: (a) magnitude and (b) phase.

reduced by the ODFC from the initial maximum 120° to approximately 20° . The time history random force waveform tracking improvement with the ODFC can be directly seen from Fig. 20, where experimental results indicate that the ODFC enhances the force tracking accuracy of the EHFS in contrast to the PI controller.

4.5. Experimental results of the proposed controller

4.5.1. Random force tracking experiment

The proposed controller can be established by incorporating the online adaptive controller with the ODFC and its experimental MATLAB/Simulink model is demonstrated in Fig. 12. To experimentally test the effectiveness of the proposed controller for the EHFS, the length and initial values of the adaptive filters are selected to be eight and zero, respectively, and the conventional PI controller, the ODFC, and the direct adaptive controller (DAC) based on the PI-controlled force closed-loop system without the ODFC are chosen as comparative controllers for the proposed controller. The first input force signal of the comparative experiments for the above mentioned four controllers is a random force input signal with an amplitude of 1500 N and a frequency range of 0–20 Hz. The adaptive online tuning process with the random input force signal for the proposed controller is shown in Fig. 21, where the time domain waveforms of the force signals are shown in Fig. 21(a) and its tracking error is shown in Fig. 21(b). Fig. 21(c) shows the normalized misalignment curves of the proposed control strategy for the EHFS using a mean squared error (MSE) index. From Fig. 21, it can be noted that the proposed control strategy yields a smaller force tracking error than the PI controller. As can be inferred from the three figures, at the beginning of the adaptive tuning process, the force tracking error is extremely large. After a period of online adaption, the force tracking error is greatly attenuated, and finally when the adaption is finished, the force tracking error remains relatively small and at a stable level compared with that at the beginning of the adaption, which indicates convergence of the adaptive algorithm.

Similar to Fig. 21, Fig. 22 presents the adaptive tuning process and its tracking error for the EHFS system using the DAC controller. It can be obviously observed that for the DAC algorithm without the ODFC, the final well-tuned random force tracking performance is greatly degraded compared with that of the proposed controller. This is due to the fact that the ODFC is employed in the proposed online adaptive controller and the phase delay problem is greatly alleviated by the ODFC, as shown in Fig. 19, which creates beneficial conditions for the subsequent adaptive compensation. However, for the DAC algorithm, there exists great phase delay error for the PI-controlled EHFS, and the phase delay error affects the adaptive tuning process, resulting in an unsatisfactory force tracking performance. The random force tracking waveform by the proposed controller when the adaption controller converges to its optimal solution, together with experimental results controlled by the PI controller, the ODFC, and the DAC are depicted in Fig. 23(a)–(d), respectively, and the force tracking errors with different controllers are shown in Fig. 23(e). Frequency response characteristics with different controllers are also presented in Fig. 24 for evaluation of different controllers. As intuitively observed from these figures, force tracking accuracy of the EHFS with the PI controller and the DAC is worse than those with the other two controllers, which results from the larger phase delay error property, as clearly indicated in Fig. 24(b). When the ODFC is applied, the phase delay error shown in Fig. 24(b) is effectively reduced and the force tracking accuracy is significantly enhanced. Finally, the best force tracking performance is achieved by the proposed controller with both little amplitude tracking error shown in Fig. 24(a) and phase tracking error shown in Fig. 24(b).

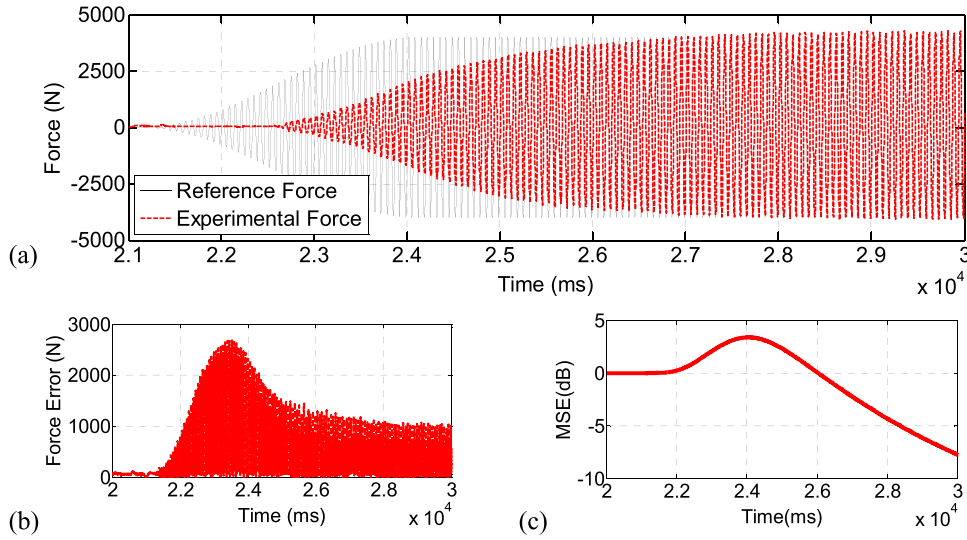


Fig. 25. Experimental results of the sinusoidal force input signal with the proposed controller: (a) adaptive tuning process and (b) its tracking error, and (c) its MSE.

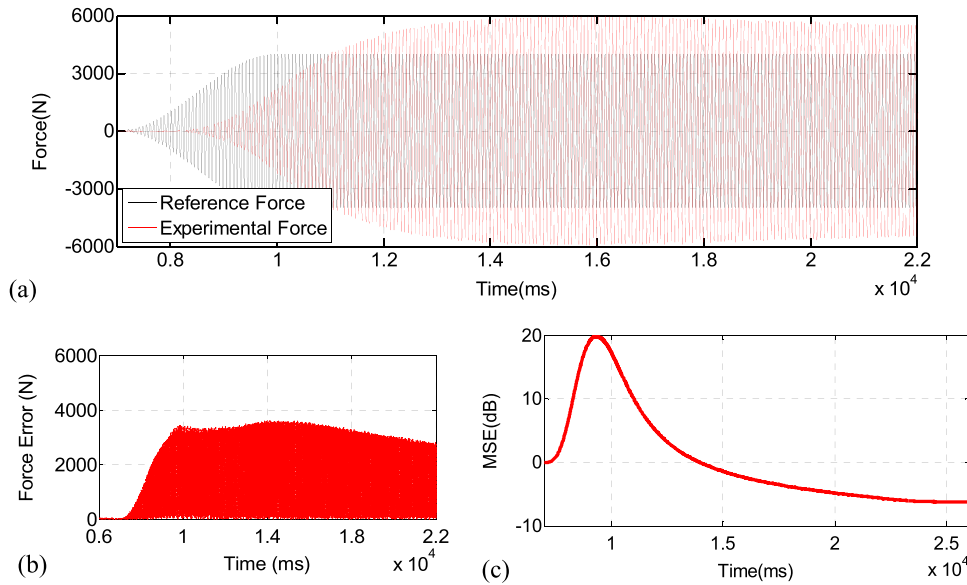


Fig. 26. Experimental results of the sinusoidal force input signal with the direct adaptive controller: (a) adaptive tuning process and (b) its tracking error, and (c) its MSE.

4.5.2. Sinusoidal force tracking experiment

In addition to the random force input signal, a sinusoidal force signal with an amplitude of 4000 N and a frequency of 15 Hz is also employed in the experiments. The adaptive tuning process for sinusoidal force tracking with the proposed controller and its adaptive tracking error are shown in Fig. 25. The adaptive tuning process with the DAC is also presented in Fig. 26 for comparison purposes. As can be concluded from these figures, the adaption process with the sinusoidal force input signal is in accordance with that for the random input force, and after a period of self-adaption, the proposed controller gradually converges, and a relatively stable and accurate tracking performance is obtained. The converged sinusoidal force tracking performance together with those with the other three controllers are shown in Fig. 27. It can be observed from Fig. 27 that the conventional PI controller and the DAC has the worst tracking accuracy, the force tracking performance is improved to some extent with the ODFC, and the highest force tracking precision is achieved with the proposed controller compared with those with the other three controllers. Therefore, the effectiveness of the proposed controller is validated.

4.5.3. Quantitative analysis of different controllers

To quantitatively compare the effectiveness of different controllers, a typical quantitatively experimental evaluation criterion using a relative root-mean-square error (RMSE) index is employed, and the RMSE index can be expressed by

$$RMSE = \sqrt{\frac{\sum_{k=1}^M [r_F(k) - y_F(k)]^2}{\sum_{k=1}^M r_F^2(k)}}, \quad (41)$$

where M is the length of reference and output signals. The RMSE index for different controllers with the above mentioned random and sinusoidal input forces are listed in Table 3. As can be seen from the table, the RMSE index for the PI controller, the ODFC, the DAC, and the proposed controller are 69.34%, 22.04%, 74.01%, and 17.13% with the random force input, and 90.66%, 28.78%, 74.16%, and 16.40% with the sinusoidal force input, respectively. Thus, the RMSE results also reveal the effectiveness of the proposed controller.

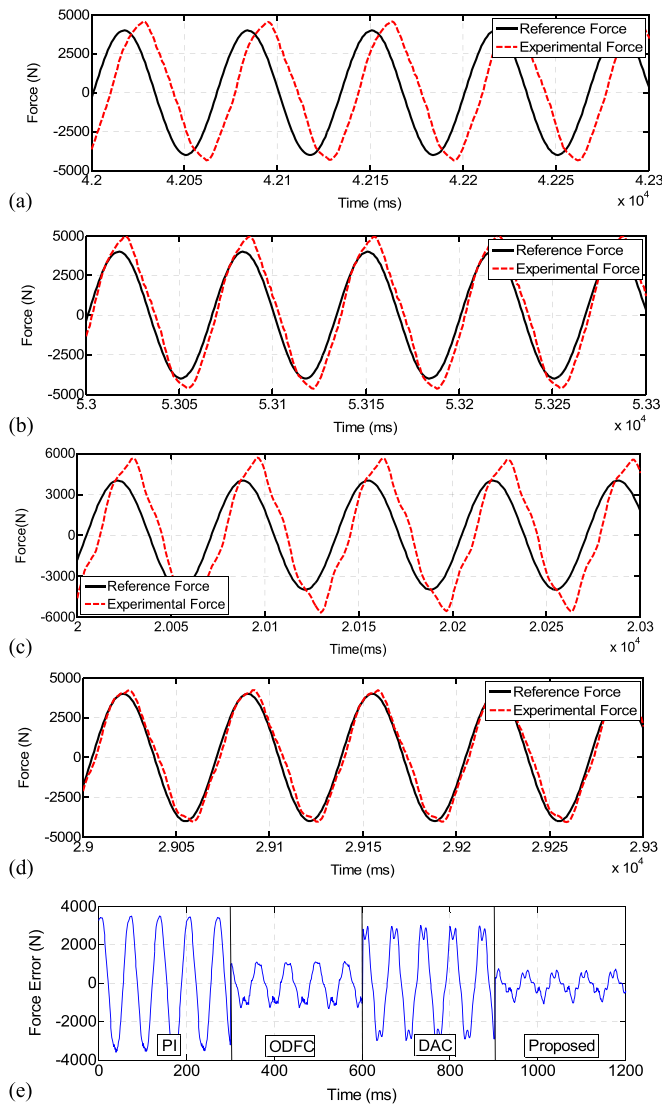


Fig. 27. Experimental results of sinusoidal force input signals with (a) the PI controller, (b) the ODFC, (c) the DAC, and (d) the proposed controller, and (e) their tracking errors.

Table 3
Comparison of the RMSE index (%) with different force input signals for different controllers.

Controllers	Random	Sinusoidal
PI	69.34	90.66
ODFC	22.04	28.78
DAC	74.01	74.16
Proposed controller	17.13	16.40

5. Conclusion

A novel controller combining an ODFC with an online adaptive compensator is proposed in this work for force tracking in the EHFS, and a CSIA is employed for identification of the force closed-loop system. An ODFC is designed by the H_∞ theory to expand the force closed-loop frequency bandwidth of the EHFS. To cope with the time-varying dynamics of the EHFS, an online adaptive compensator combined with the ODFC is proposed. Experiments are conducted on an xPC rapid prototype to validate the practical feasibility of the proposed controller. Experimental results indicate that a better force tracking performance improvement is obtained

compared with the PI controller, the ODFC, and the direct adaptive controller.

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References

- [1] Plummer AR. Control techniques for structural testing: a review. *IMechE I: J Syst Control Eng* 2007;221(2):139–69.
- [2] Shen G, Zhu ZC, Zhang L, et al. Adaptive feed-forward compensation for hybrid control with acceleration time waveform replication on electro-hydraulic shaking table. *Control Eng Pract* 2013;8(21):1128–42.
- [3] Shen G, Zhu ZC, Li X, et al. Real-time electro-hydraulic hybrid system for structural testing subjected to vibration and force loading. *Mechatronics* 2016;33:49–70.
- [4] Kim JW, Xuan DJ, Kim YB. Design of a forced control system for a dynamic road simulator using QFT. *Int J Automat Technol* 2008;9(1):37–43.
- [5] Plummer AR. Model-based motion control for multi-axis servohydraulic shaking tables. *Control Eng Pract* 2016;53:109–22.
- [6] Yao JY, Jiao ZX, Ma DW, Yan L. High-accuracy tracking control of hydraulic rotary actuators with modeling uncertainties. *IEEE/ASME Trans Mechatron* 2014;19(2):633–41.
- [7] Wang CW, Jiao ZX, Wu S, Shang Y. Nonlinear adaptive torque control of electro-hydraulic load system with external active motion disturbance. *Mechatronics* 2014;24(1):32–40.
- [8] Yang CF, Qu ZY, Han JW. Decoupled-space control and experimental evaluation of spatial electro-hydraulic robotic manipulators using singular value decomposition algorithms. *IEEE Trans Ind Electron* 2014;61(7):3427–38.
- [9] Yao JY, Jiao ZX, Ma DW. Extended-state-observer-based output feedback nonlinear robust control of hydraulic systems with backstepping. *IEEE Trans Ind Electron* 2014;61(11):6285–93.
- [10] Li XC, Chen SL, Teo CS, Tan KK, Lee TH. Data-driven modeling of control valve stiction using revised binary-tree structure. *Ind Eng Chem Res* 2015;54(1):330–7.
- [11] Riccardo BC, Claudio S, Gabriele P. System identification applied to stiction quantification in industrial control loops: a comparative study. *J Process Control* 2016;46:11–23.
- [12] Brasio AS, Romanenko A, Fernandes NC. Modeling, detection and quantification, and compensation of stiction in control loops: the state of the art. *Ind Eng Chem Res* 2014;53:15020–40.
- [13] Kim WH, Shin DH, Won DH, Chung CC. Disturbance-observer-based position tracking controller in the presence of biased sinusoidal disturbance for electrohydraulic actuators. *IEEE Trans Control Syst Technol* 2013;21(6):2290–8.
- [14] Niksefat N, Sepehri N. Designing robust force control of hydraulic actuators despite system and environmental uncertainties. *IEEE Control Syst Mag* 2001;21(2):66–77.
- [15] Nam Y, Hong SK. Force control system design for aerodynamic load simulator. *Control Eng Pract* 2002;10(5):549–58.
- [16] Ahn KK, Dinh QT. Self-tuning of quantitative feedback theory for force control of an electro-hydraulic test machine. *Control Eng Pract* 2009;17(11):1291–306.
- [17] Baghestan K, Rezaei SM, Talebi HA, et al. Robust force control in a novel electro-hydraulic structure using polytopic uncertainty representation. *ISA Trans* 2014;53(6):1873–80.
- [18] Sohl GA, Bobrow JE. Experiments and simulations on the nonlinear control of a hydraulic servosystem. *IEEE Trans Control Syst Technol* 1999;7(2):238–47.
- [19] Nakkarat P, Kuntanapreeda S. Observer-based backstepping force control of an electrohydraulic actuator. *Control Eng Pract* 2009;17(8):895–902.
- [20] Yao JY, Jiao ZX, Yao B. Nonlinear adaptive robust backstepping force control of hydraulic load simulator: theory and experiments. *J Mech Sci Technol* 2014;28(4):1499–507.
- [21] Wu G, Sepehri N, Ziaei K. Design of a hydraulic force control system using a generalised predictive control algorithm. *IEE Control Theory Appl* 1998;145(5):428–36.
- [22] Tang Y, Zhu ZC, Shen G. Design and experimental evaluation of feedforward controller integrating filtered-x LMS algorithm with applications to electro-hydraulic force control systems. *IMechE C: J Mech Eng Sci* 2016;230(12):1951–67.
- [23] Yao JY, Jiao ZX, Shang YX, et al. Adaptive nonlinear optimal compensation control for electro-hydraulic load simulator. *Chin J Aeronaut* 2010;23(6):720–33.

- [24] Gao BW, Shao JP, Yang XD. A compound control strategy combining velocity compensation with ADRC of electro-hydraulic position servo control system. *ISA Trans* 2014;53(6):1910–8.
- [25] Wang CW, Jiao ZX, Quan L. Nonlinear robust dual-loop control for electro-hydraulic load simulator. *ISA Trans* 2015;59:280–9.
- [26] Truong DQ, Ahn KK. Force control for hydraulic load simulator using self-tuning grey predictor – fuzzy PID. *Mechatronics* 2009;19(2):233–46.
- [27] Zhao JS, Shen G, Zhu WD, Yang CF, Yao J. Robust force control with a feed-forward inverse model controller for electro-hydraulic control loading systems of flight simulators. *Mechatronics* 2016;38:42–53.
- [28] T.L. Lin, W.P. Huang, H.L. Ren, et al., New compound energy regeneration system and control strategy for hybrid hydraulic excavators. *Automat Constr*, 68, 2016, 11–20.
- [29] Y. Liu, G.F. Gong, H.Y. Yang, D. Han, H. Wang, Regulating characteristics of new tamping device exciter controlled by rotary valve. *IEEE/ASME Trans Mechatron*, 21(1), 2016, 497–505.
- [30] Vaes D, Engelen K, Anthonis J, et al. Multivariable feedback design to improve tracking performance on tractor vibration test rig. *Mech Syst Signal Proc* 2007;21(2):1051–75.
- [31] Langdon R, Docherty PD, Chiew YS, et al. Implementation of a non-linear autoregressive model with modified Gauss-Newton parameter identification to determine pulmonary mechanics of respiratory patients that are intermittently resisting ventilator flow patterns. *IFAC-Pap OnLine* 2015;48(20):354–9.
- [32] Vaudrey MA, Baumann WT, Saunders WR. Stability and operating constraints of adaptive LMS-based feedback control. *Automatica* 2003;39(4):595–605.